CMPE 484

Mobile Application Project Proposal

Team Name: Paginex

Team Members: Mert Bölükbaşı 20221701046
Burak Özevin 20231701014
Ramazan Birkan Öztürk 20221701025
Suhan Arda Öner 20221701059
Elif Zeynep Talay 20231701024

1. Introduction

- **Problem Definition:** Even though multiple people read books, there is a scarcity in social platforms where people can interact with each other. Most people prefer to observe which book is popular, what type of book their friends read, or new books that are worth reading on a social media platform. For example, most people try to share posts about the books that they read on Facebook, Instagram, or Reddit, but these posts are not organized in a book-focused structure. That's why the interaction is restricted and distributed. This situation causes some difficulties in sharing experience and information about books between users and prevents the enhancement of the reading culture.

- **Proposed Solution:** To solve the problem, we proposed a book social media application. This application provides an interactive platform that users can follow, share, evaluate, and comment on the books. The users can follow each other, get book suggestions according to their area of interests, create booklists, and share their book tastes with other users. The application aims to enhance the social interaction and the reading experience through basic components such as user management, scoring and comment modules, and community features. Thus, users can get data and feedback about the books they read, and can enlarge their networks within the book community.

- **Target Audience:** The primary target audience of this application is people who like to read books and want to create networks about the books. The target audience consists especially:
  - Students and young adults
  - The members of book clubs and groups
  - The ones who regularly read books and likes to find new books
  - Social media users who actively use the technology

  The target audience has high motivations for exchanging their book reviews, finding new masterpieces, networking with those who have similar book tastes, and creating user profiles that will benefit most from the application.

2. Application Overview

- **Core Features:** In the application, there will be three main pages: main page (feed), explore, and profile. In order to view the content of the application and interact with the application, users must register or login.
  - **Account and User Management**: Users can register and login (authentication system). They can create their profiles and change their profile information.
  - **Book Sharing**: Users can share the books they read. The content of the share includes the book title, genre, author, summary, and the user's review.
  - **Reading Status:** Users can choose reading status for a book: Read, Reading, On-Hold (Temporary), Dropped (Permanent), Want-to-read. This status will be shown on the post and the profile of the user.
  - **Interaction:** Users can like the posts. Comments can be written to the posts, and current comments can be viewed.
  - **Save:** Users can save the posts in order to see the post again. Saved posts are listed in the archive of the user.
  - **Following System:** Users can follow or unfollow other users. The followed users will be shown on the main page.
  - **Explore:** Users uses Explore page to view book and book list posts. The posts are

listed according to users' areas of interest.

- **Book Lists:** Users can create book lists and share them. Other users can view and follow these book lists. Users can also add these to their profiles.
- **Mind Map:** According to the shared books of the user, a mind map will be created. The mind map is automatically created as a User-Genre-Book structure. The nodes are derived from the posts of the user.

- **Technical Approach:** The application will be developed by using Kotlin programming language and Android Studio. For the authentication operations, Firebase Authentication will be used. Posts, comments, likes, saves, book lists, and reading status are stored in Cloud Firestore. Data security and access control will be provided by Firebase Security Rules.

3. Project Scope and Objectives

- **Project Scope:** In this project, we will use an authentication system with Firebase. Users can create and edit their own profiles, share books, like, comment on, and save the shared books, follow each other, reach other users through the explore page, and create and follow reading lists. The project will not include book purchasing, real-time chat, a notification system, or multi-platform support.

- **Project Objectives:** The objective of this application is to create an Android-based social media platform in which the readers can interact with each other. Within the scope of the application, a secured authentication system and an adjustable profile system will be developed. We aim to create features such as book sharing, a like-comment system for posts, post saving, unfollow-follow system for users. Also, we aim to create Explore UI in which users can find new books to read, a booklist creation and sharing system, unfollow-follow system for booklists. Finally, the goal is secure the conservation of all data in Firebase and propose a working prototype with basic features by the end of the project.

4. Team and Responsibilities

- **Team Members:**
  1. Mert Bölükbaşı: Backend & Database Developer
  2. Burak Özevin: Backend & Security Developer
  3. Ramazan Birkan Öztürk: Frontend Developer
  4. Suhan Arda Öner: Frontend Developer
  5. Elif Zeynep Talay: Frontend Developer

- **Responsibilities:**
  - Mert Bölükbaşı: Design of database model, CRUD operations and data integrity, secure management of the data
  - Burak Özevin: Authentication, authorization, Firebase security rules, input validation, basic abuse prevention
  - Ramazan Birkan Öztürk: Profile UI (profile viewing, profile adjustment)
  - Suhan Arda Öner: Design of Explore UI
  - Elif Zeynep Talay: Design of the main page of the application, the design of mind map

5. Conclusion

The proposed Paginex application targets a social media platform for book readers. The proposed application is primarily focused on the reading habit and facilitates more meaningful knowledge and experience sharing by shifting book-related activities from general social media to a book-centric environment. The proposed application helps readers connect, share content, and create their own reading networks. We are developing an app using Kotlin and Firebase that is secure and allows users to easily find, review, and share their favorite books. As a conclusion, Paginex helps readers connect, transforming a solo activity into a social community.